

Erwan MARTIN

Available from **July 2026** for a **Data Scientist / Machine Learning Engineer** position

📍 Paris, France | ✉ erwan2309@gmail.com | ☎ +33 6 22 79 69 15 | 🔗 LinkedIn | 🐙 Github

Profile

Data Scientist with applied Machine Learning experience across R&D, industrial and healthcare environments. Skilled in supervised & unsupervised learning, deep learning, signal processing and large-scale data pipelines (Python, TensorFlow, PyTorch, Scikit-Learn, XGBoost, MLflow, SQL, AWS, HPC). Comfortable in multinational, multicultural and English-speaking environments (TOEIC 910/990).

Professional Experience



R&D Data Scientist (fixed-term contract), TotalEnergies – Pau, France 🇫🇷

July 2025 – July 2026
1 year

Project: design an AI solution for marine biodiversity monitoring from Distributed Acoustic Sensing (DAS) data.

- Built a reference acoustic database of marine species through systematic scientific literature review
- Performed signal processing on **53 TB** of raw data on **HPC clusters (Slurm & LSF schedulers)**: **Butterworth/F-K** filtering, energy computation and spectrogram extraction (**FFT**) in **Python**, parallelised with **Dask** (**-89%** compute time)
- Trained **convolutional auto-encoders (CNN)** for **feature extraction** on spectrograms (**TensorFlow**)
- Applied **unsupervised clustering (K-Means, HDBSCAN)** on learned embeddings for species identification
- Localised acoustic sources via **geometric optimisation** on (x, y, z) coordinates with **uncertainty quantification**



Data Scientist / Android Developer Intern, Aneo – Boulogne-Billancourt, France 🇫🇷

May 2024 – November 2024
6 months

Project: finalise an Android liver-spectrometry application to estimate macro-steatosis from patient data, in partnership with the BOPA innovation chair (AP-HP, Institut Mines-Télécom, Université Paris-Saclay).

- Finalised an **Android** liver-spectrometry application (**React, Ionic, TypeScript**)
- Built a **CI/CD** pipeline (**Python**) and **cloud storage** layer (**AWS S3, AWS Lambda**)
- **Deployed** the application to production for clinical data collection at Paul Brousse Hospital, in collaboration with surgical teams
- Developed and tracked Machine Learning models with **Kedro** and **MLflow** (**regression, Random Forest, SVM**)



Data Scientist Intern, Dow – Tarragona, Spain 🇪🇸

Sep. 2023 – Mar. 2024
6 months

Project: develop a Machine Learning model predicting the mechanical properties of plastic films from their chemical composition.

- **Extracted, cleaned and integrated** experimental data (**SQL, Python, Pandas, NumPy**)
- Benchmarked **supervised models: Deep Learning, XGBoost** (**TensorFlow, scikit-learn**)
- **Deployed** two production **XGBoost** models ($R^2 = 0.94$ and 0.82) through a **Streamlit** application for industrial use



Data Scientist Intern, Thales – Vélizy-Villacoublay, France 🇫🇷

May 2022 – September 2022
4 months

Project: develop a Machine Learning model to predict the trajectory of an electron beam from electron-gun parameters (aerospace & defence R&D).

- **Monte Carlo simulation** (**ANSYS, Persalys**) and synthetic data generation
- Built **AutoML** and **Deep Learning** models (**H2O, TensorFlow**) for trajectory prediction

Education

CY TECH (formerly E.I.S.T.I) – Engineering degree (Master's level) in Applied Mathematics, specialisation in Data Science

Sep. 2020 – Nov. 2024

- Core modules: Machine Learning • Deep Learning • Probability • Statistics • Time Series • Graph Theory • Optimisation
- Parallel Programming • Signal Processing • NLP • Constraint Programming

Lycée Jean-Baptiste Corot – French CPGE (intensive preparatory programme), Mathematics & Physics track (MPSI – MP)

Sep. 2018 – Jul. 2020

Skills

Languages: French (Native), English (Advanced, C1, TOEIC 910/990), German (Beginner, A2)

Machine Learning & AI: Supervised, unsupervised & self-supervised learning • Deep Learning (CNN, auto-encoders) • Clustering (K-Means, HDBSCAN) • XGBoost • Random Forest • SVM • AutoML • NLP • Optimisation • Statistical testing & regression

Programming: Python (primary) • R • SQL • TypeScript | **Python ecosystem:** NumPy • Pandas • Scikit-Learn • TensorFlow • PyTorch • XGBoost • NLTK • Librosa • Dask • Kedro • MLflow • Selenium

Databases: MySQL • MariaDB • Oracle • AWS S3 | **Cloud & DevOps:** AWS (S3, Lambda) • Git • Azure DevOps • CI/CD • HPC clusters (Slurm, LSF)

Visualisation: Streamlit • Plotly • Dash • Matplotlib • Seaborn • Power BI • SAS | **Methodology:** Agile / Scrum